Random Colored Line Hierarchy Task (RCLHT):

Parallel and Serial Accumulation of Information in Decision Making

Honors Thesis

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1 Acknowledgement

First and foremost, I would like to thank the Laboratory for Neural Computation and Cognition (LNCC) team. When I first joined the lab, I had barely decided on my concentration, and knew next to nothing about the field of cognitive neuroscience or research in general. I have learned so much in the past two years, and I truly appreciate all the help and guidance offered by the members of LNCC. Specifically, I would like to thank Professor Michael Frank for so kindly welcoming me to his lab and project. I would also like to thank Dr. Matt Nassar for his continuous guidance through every step of the study, from understanding literatures to designing and analyzing for the project. I cannot express how grateful I am for all his time and the incredible amount of patience he's had with me in the past year and a half.

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3. Introduction

3.1 Decision Making and Sequential Analysis

A decision is a process that involves deliberation of likelihoods (contemplating on which hypothesis most likely caused the observed state) and commitment to a categorical proposition (the act of making decision itself) (Gold 2007). Not all decisions have explicit answers, however, and in such cases, the deliberation process would involve weighing of values in alternative actions, represented by costs and benefits of the corresponding outcomes. From daily lunch menu selections to marriage that lasts a life time, decision making is a considerable part of everyone's life, and is thus subject to investigation in various fields of study, such as neuroscience, psychology, statistics, and economics (Gold 2007). Most work in neuroscience focuses on simple binary decisions made in sensory-motor tasks, as they allow for a better control of input stimulus and clearer observation of responsive brain activities. For example, several past studies with sensory-motor tasks observed the lateral intraparietal area (LIP), superior colliculus (SC), and frontal eye field (FEF) to be some of the decision-related areas in brain (Roitman & Shadlen 2002, Horwitz & Newsome 2001, Ding & Gold 2012, Ferrera et al. 2009).

Mathematically, simple decision-making processes can be described as a form of statistical inference, where the likelihood of a hypothesis is updated with evidence, or relevant information obtained (Gold 2007). In the case of sensory-motor tasks, evidence may be represented by a single neuron, a pool of neurons, or the relationship between their spike rates (Gold 2007). One mathematical framework for studying such a process over time is sequential analysis, which continuously updates the likelihood of a hypothesis as an unlimited sequence of evidence is obtained. Sequential analysis thus naturally addresses the issue of speed-accuracy trade-off that pertains to every decision-making process (Forstmann 2016). Because the process of making a decision is always under some kind of time constraint, there is an inevitable trade-off between collecting more evidence for a greater likelihood of a single hypothesis and ceasing collection and making do with the existing likelihood values. One of the strategies that deal with this issue is Wald's Sequential Probability Ratio Test (SPRT), which optimizes the performance of a binary decision for a given accuracy (Boagacz et al. 2006). SPRT is widely used especially in the field of neuroscience because of its optimized efficiency, which is thought to well represent neurons that have evolved to process information with great efficiency (Ma et a. 2006).

3.2 Random Dot Motion and Drift Diffusion Model

The Random Dot Motion (RDM) paradigm is widely used as a sensory-motor task in the field of neuroscience. For RDM tasks, a subject's goal is to identify the direction of coherently moving dots among other dots with random movements. The difficulty of the task is determined by changing the amount of coherently moving dots. For instance, when implementing this task in primates, the monkeys would indicate the direction they believe a majority of the dots are moving by making saccade to the corresponding direction.

Behavior in the RDM paradigm is well accounted for by the Drift Diffusion model (DDM), which is a type of sequential sampling models that utilizes a relative evidence rule and deals with continuous evidence accumulation (Forstmann 2016). DDM allows for manipulation of four key parameters (drift rate, boundary separation, starting point bias, and non-decision time) and three across-trial variabilities (in drift rate, starting point bias, and non-decision time) (Ratcliff et al. 2016).

3.3 Hierarchical Representation of Conjunctive Features

While a big proportion of the current and past studies focuses on the simple binary decision making processes, there are increasing interests in studies of more complex decisions, such as one involving multiple alternatives (McMillen & Holmes 2006) or hierarchical reinforcement learning (Botvinick 2012). In the recent years, several studies have reported evidence that information and cognitive processes may be hierarchically organized in the prefrontal cortex (PFC), which would allow one to perform well in a hierarchically-structured environment (Frank & Badre 2011). A hierarchically-structured task would involve observation of multiple features of a stimulus and implementation of more than one task-set, where observation of one feature would determine the appropriate task-set to be implemented in a trial. For example, one may learn to (or be instructed to) respond differently to the color of a stimulus depending on its shape (i.e. press ‘A’ if yellow and ‘B’ if green for a circular stimulus but press ‘B’ if yellow and ‘A’ if green for a triangular stimulus). In order to respond correctly for such a task, one could first pay attention to the shape of the given stimulus and determine whether it is circular or triangular. Then, the corresponding task-set (‘A’-yellow, ‘B’-green; or ‘A’-green, ‘B’-yellow) would be considered to determine the appropriate action. However, it is also possible that the processing of evidence for shape and color is not as distinct, or serial, as the case above. Some evidence for shape and color may also be collected simultaneously, or in a parallel manner. Although not explicit, evidence for both methods is observed in some hierarchical reinforcement learning studies (Collins & Frank 2013).

Our task was designed to investigate the hierarchical decision making process. We organized three features into a hierarchical structure of two dimension levels, each requiring a simple binary decision. Prior to the study, we considered four different possibilities of information processing methods: (1) complete parallel processing, (2) complete serial processing, (3) compressed parallel processing, and (4) compressed serial processing. The first two complete processing methods would accumulate information on all presented features (three in our task) before committing to a decision, while the last two compressed processing methods would accumulate information only on the two relevant features for the decision. Parallel processing methods would involve accumulating information on all features simultaneously, while serial processing methods would involve going through each feature one by one in terms of decision-making.

Our goal in this study is to develop a paradigm with which we could dissociate serial from parallel processing. In order to do so, we created a hierarchical random dot motion task and developed a statistical approach that parameterizes the extent to which an individual utilizes a serial or parallel processing scheme, and whether individuals adapt their processing strategy as they learn the task. Our analysis for the study focused on the compressed serial and parallel processing methods.

4. Materials & Methods

4.1 Participants

A total of 49 subjects (20 male, 29 female) between the age of 18 and 26 (M = 19.92, SD = 2.02) participated in the study. Of the 49 subject data, 13 were excluded from analysis due to incompleteness of data collection or inconsistency in parameter selection. The remaining 36 subject data (18 with a more difficult coherence set, 18 with an easier coherence set) were used in our statistical analysis.

Participants were recruited through advertisement flyers on Brown University campus or SONA, a research participation system for Cognitive, Linguistic & Psychological Studies (CLPS) department at Brown University. Of the 49 subjects total, 19 were compensated at $10 per hour while the remaining 30 were given course credits for their contributed time. For 33 out of 49 subjects, our task for the study was paired with a task for another study at the Laboratory for Neural Computation and Cognition (LNCC) to form a two-hour-long study in total. Of these 33 subjects, 20 carried out our task in the first hour, while the remaining 13 carried out the task in the second hour. For each subject, demographic survey and informed consent was obtained in a manner approved by the Brown University Institutional Review Board.

4.2 Experimental Procedure

Random Colored Line Hierarchy Task (RCLHT) was developed as a variation of random dot motion task and was designed to measure the categorical perception of three orthogonal features: (1) movement direction, (2) color, and (3) angle. The task was carried out in a darkened room and consisted of 512 trials split into four blocks, between which a subject could take a short break for eye relaxation. In each trial, a large number of colored lines were shown on screen (91 per video frame), and subjects were required to respond by pressing one of four keyboard keys: 'D', 'F', 'J', or 'K'. After each response, a feedback image (thumbs-up or thumbs-down) was shown for 0.6 seconds followed by 1 second of blank screen with fixation point before displaying a new set of lines for the next trial.

Each of the three features involved two states—right or left for movement direction, yellow or pink for color, and negative or positive slope for angle. For each trial, some fraction of the lines on screen moved to a unified direction of left or right, and each of the lines were colored either yellow or pink and were angled either with positive or negative slope. For color and angle features, subjects were instructed to identify the dominant state. The three features were used to form a hierarchical structure of two dimension levels, such that there are one higher dimension feature and two lower dimension features. In each trial, the state of the higher dimension feature would determine the relevant lower dimension feature of which information is required to respond correctly. The combination of features for the hierarchical structure was counterbalanced across subjects, and a complete counterbalanced set involved six tasks, each with different hierarchical combination of the three features.

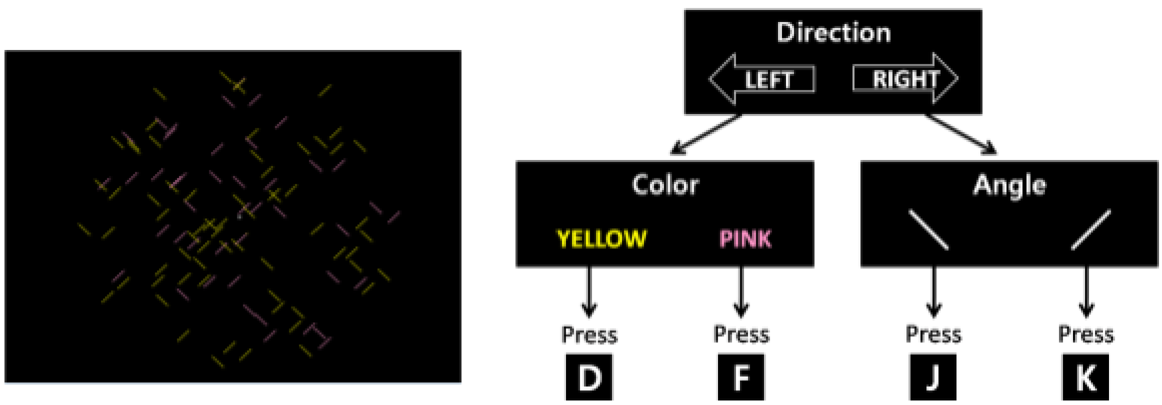

Figure 1. RCLHT Task Design
(A) Screenshot of an example trial (a single video frame).
(B) An example hierarchical structure is formed with movement direction as the higher dimension feature and color and angle as the lower dimension features. Depending on the state of the movement direction (left or right), either color or angle will serve as the relevant lower dimension feature required to respond correctly.

Two different coherence levels (high and low) were used to increase or decrease the difficulty of decision making for each feature. High coherence level for movement direction involved a large percentage of lines moving in a unified direction, while the rest moved in random directions. Low coherence level involved a smaller percentage of lines moving in a unified direction, while the rest moved in random directions. High coherence level for color involved a large percentage of lines colored in the dominant color, while low coherence level involved smaller percentage of lines (but greater than 50%) colored in the dominant color. Likewise, high coherence level for angle involved a large percentage of lines in the dominant angle, while low coherence level involved a smaller percentage of lines (still greater than 50%) in the dominant angle. Each subject participated in one of two versions of the task, either with a relatively higher coherence set (task more difficult) or with a relatively lower coherence set (task easier).

	Version 1 Coherence Levels		Version 2 Coherence Levels	
	High	Low	High	Low
Movement Direction	0.4	0.2	0.6	0.3
Color	0.7	0.58	0.72	0.63
Angle	0.7	0.58	0.72	0.63

Table 1. Coherence levels for the two versions of the task
(A) The coherence levels for version 1 were selected by fitting a psychometric function on the accuracy of binary decisions for each feature, obtained from 4 pilot data set. Pilot data tested only a feature per trial, and we roughly aimed for 95% accuracy for high coherence level and 70% accuracy for low coherence level in each feature.
(B) For version 2, all coherence levels were increased in general to obtain a higher accuracy, and thus a greater number of correct trials to analyze.

Before the actual task, all subjects were given practice trials with easier coherence levels to get acquainted with making binary decisions for the three features independently (10 trials for each feature) and collectively (3 sets of 10 trials). To ensure that the hierarchical combination rules were learned, subjects were required to get 90%

or higher accuracy on the third set of practice trials before moving on to the actual task. A set of 10 practice trials was repeated until the 90% accuracy was obtained.

4.3 Statistical Analysis

Our analysis focused on the compressed parallel and compressed serial processing models, in which only the two relevant features (one higher dimension feature and one lower dimension feature) were taken into account when measuring response time for each trial. We predicted that the two processing methods would be distinguishable by the relative response time of the four coherence level combinations. From here on, we will refer to the four coherence level combinations as HH, HL, LH, and LL, where 'H' refers to high coherence and 'L' refers to low coherence. For each combination of two letters, the first letter represents the coherence level of the higher dimension feature and the second letter represents the coherence level of the relevant lower dimension feature. Our intuition is described in Figure 2. (See Appendix: Mathematical Proof of Intuition for the mathematical logic behind this intuition.)

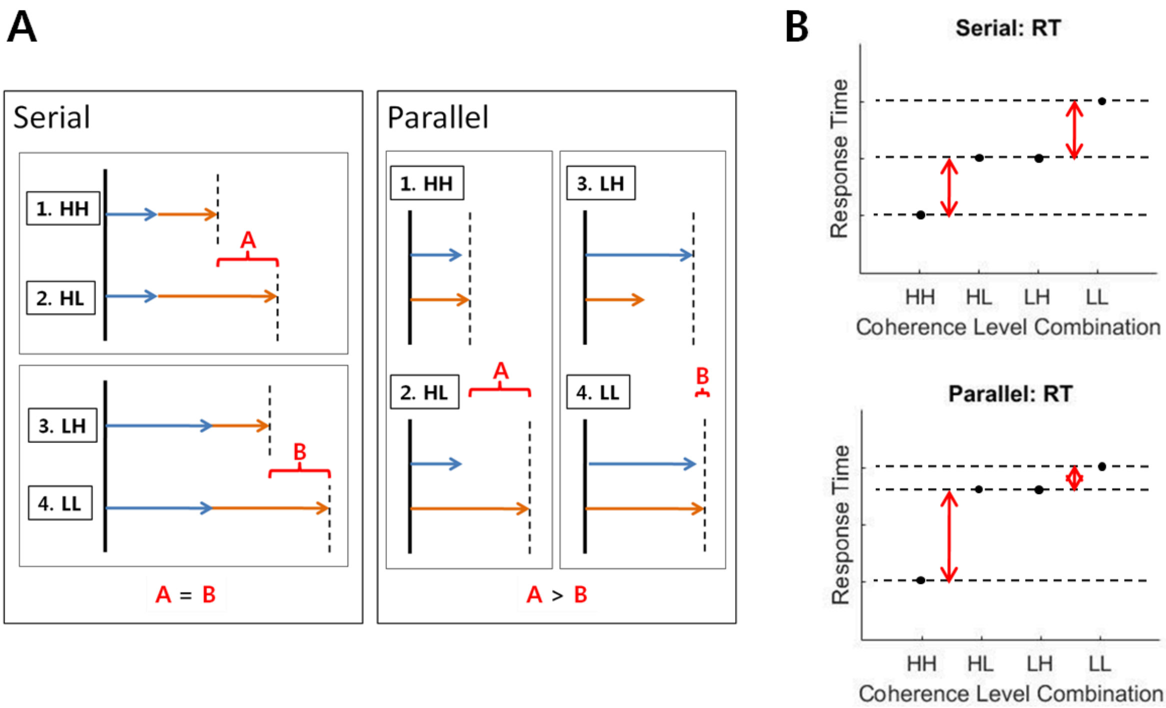

Figure 2. Distinguishing Parallel and Serial Processing Methods.
(A) Let the blue arrows represent the response time of the higher dimension feature and the orange arrows represent the response time of the relevant lower dimension feature. Intuitively, a high coherence level leads to a short arrow and a low coherence level leads to a long arrow. In the serial processing model, a pair of blue and orange arrows would be added in order to represent the overall response time. In the parallel processing model, the longer of the two arrows will be considered equivalent to the overall response time. We can notice the different relationship between the red-lettered A and B for the serial and parallel processing methods. For parallel processing method, the lower dimension feature's coherence level will contribute more to the overall response time when the higher dimension feature's coherence level is high than when it is low (A > B). This relationship does not hold for the serial processing method (A = B).
(B) Our intuition is demonstrated in the form of RT plots. We expect the response time for the interacting coherence combinations (HL and LH) to be equally spaced between LL and HH in the serial processing model, but closer to LL than HH in the parallel processing model.

We parameterized the extent to which an individual utilizes a serial or parallel processing scheme using a linear regression model on the inverted-response time (speed) of the correctly responded trials. The regression model contained four predictor variables: coherence level of the higher dimension feature (X_1), coherence level of the relevant lower dimension feature (X_2), coherence level of the irrelevant lower dimension feature (X_3), and interaction between the coherence levels of the higher and relevant lower dimension features (X_1X_2). The predictor variables for individual features (X_1, X_2, and X_3) were set to 1 when the coherence level was high and -1 when the coherence level was low. Naturally, the interaction (X_1X_2) was set to 1 when the higher and relevant lower dimension features had identical coherence levels (HH or LL) and -1 when they were different (HL or LH). Since our task involved equal number of trials for each of the coherence level combination, all of our predictor variables were centered with mean 0.

Although our original method for distinguishing between the parallel and serial processing models was in terms of response time, we used inverted response time for the regression analysis to obtain a more normally-distributed dependent variable. With this regression model, we expect the interaction (X_1X_2) to have a greater positive coefficient with the parallel processing method than with the serial processing method. We also expect the coefficient for the irrelevant dimension feature (X_3) to equal zero, as neither of our processing models anticipate the irrelevant dimension to have influence on task performance. Given the complicated nature of our task, trials with response time shorter than 200 milliseconds were excluded from analysis as they were most likely not a result of properly-processed response.

5. Results

5.1 Simulated Performance

Simulations were developed to validate our strategy for distinguishing between the parallel and serial processing methods with the interaction term from the linear regression model described in section 4.3.

The main simulation was generated using the Ratcliff Diffusion Model, a popular sequential sampling model for the neurophysiological study of two-choice decision making. Ratcliff Diffusion Model formulates decision by accumulating noisy evidence from starting point until a boundary of decision threshold is reached (Forstmann 2016). The simulation was developed using Model Analysis Toolbox (DMAT), which allows for easy manipulation of Ratcliff Diffusion Model's seven free parameters—boundary separation, starting point, drift rate, non-decision time, and three inter-trial variabilities in (starting point) bias, non-decision time, and stimulus quality (drift rate) (Vandekerckhove & Tuerlinckx 2008). Because our task design involved two coherence levels for each feature, two parameter sets were created prior to sampling. The parameter sets used in this simulation are as follows:

	Boundary Separation	Starting Point	Drift Rate	Non-Decision Time	Variability in (Starting Point) Bias	Variability in Non-Decision Time	Variability in Stimulus Quality (Drift Rate)
1)	0.16	0.08	0.30	0.10	0	0.0001	0
2)	0.16	0.08	0.10	0.10	0	0.0001	0

Table 2. Seven Parameters in Ratcliff Diffusion Model Simulations

In order to account for the hierarchical structure in our task design, we independently generated two sets of 10,000 data points, one representing the higher dimension and the other representing the relevant lower dimension. Then, the response time for the serial processing model was computed by adding each pair of data points, while the response time for the parallel processing model was computed by taking the maximum of the two. This was analytically equivalent to generating two sets of data points for the lower dimension and then selecting one from each pair. To test our linear regression model on the simulated data, we additionally generated another set of 10,000 data points to represent the irrelevant lower dimension. The dependent variable (1/RT) was standardized prior to regression to facilitate comparison between the serial and parallel processing models. The regression results were as follows (See Figure 3): For both serial and parallel processing models, significant effects were observed for the coherence levels of the high dimension feature (serial $t(9)=87.59$, $p<0.0001$; parallel $t(9)=85.97$, $p<0.0001$) and the relevant low dimension feature (serial $t(9)=139.77$, $p<0.0001$; parallel $t(9)=139.04$, $p<0.0001$), with higher coherence leading to higher speed in general. As expected, no significant effect was observed for the coherence level of the irrelevant low dimension feature in either of the two processing models. Significant effect was observed for the interaction between the coherence levels of the high dimension feature and the relevant low dimension feature for both processing models (Serial: $t(9)=17.60$, $p<0.0001$; Parallel: $t(9)=24.83$, $p<0.0001$), with the response time of HL and LH (the interacting coherence level combinations) closer to the response time of LL than to that of HH. And as we expected, the coefficients for the interaction term were in general greater for the parallel processing model than for the serial processing model. To further examine the interaction effect and its cause, we divided all data points of each coherence combination condition into quintiles by response time and created delta plots (Figure 4). The effect was observed to be greater for the slower trials (the tail of RT distribution) in general. This may be due to the fact that the RT distribution for the harder condition (LL) has a longer tail than RT distribution for the easier condition (HH), which results in the maximized difference between the two conditions in the slower trials. For the parallel processing model, the conditions LH and HL are closer to condition LL than condition HH in distribution, and thus, the interaction effect would be most emphasized by the slower trials. This would also explain why some interaction effect was observed for the slowest quintile of the serial processing model, although to a lesser extent since the conditions LH and HL are not as similar to condition LL as in the parallel model. Intuitively, a slow trial of condition LH or HL likely indicates a relatively slow processing of the easy dimension 'H', and it is not likely that processing difficult dimension 'L' instead would take as longer since it is already very slow. Thus, the slow trials of condition LL would be closer to the slow trials of condition LH and HL than normal, demonstrating an 'interaction effect' selectively for the slower trials.

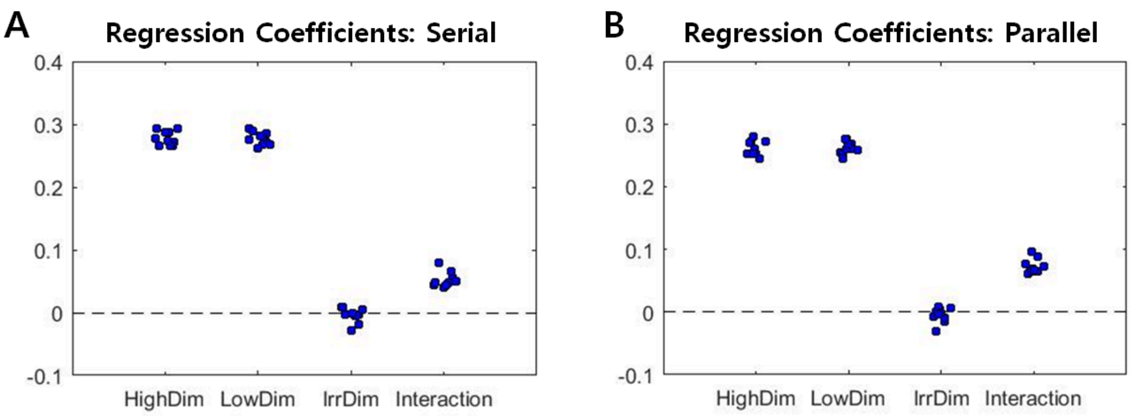

Figure 3. Regression Coefficients of Simulated Data
Simulation of 10,000 data points was repeated ten times and each of the resulting regression coefficients were plotted. X-axis is showing the regression predictor variables.
(A) For serial processing model, we observed significant effects of all predictor variables but the irrelevant dimension's coherence level.
(B) For parallel processing model, we also observed significant effects of all predictor variables but the irrelevant dimension's coherence level. The interaction coefficients seem to be slightly greater than those from the serial processing model in general.

Similar results were obtained from other simulations that we generated using a random variable with gamma distribution as reaction time. Again, two sets of 10,000 data points were generated to represent the hierarchical structure of our task. The parameters for the random variable were gamma = 2 and beta = 1. We found significant effects for all predictor variables but the irrelevant dimension feature's coherence level, and the parallel processing model's interaction term coefficient was greater than that of the serial processing model.

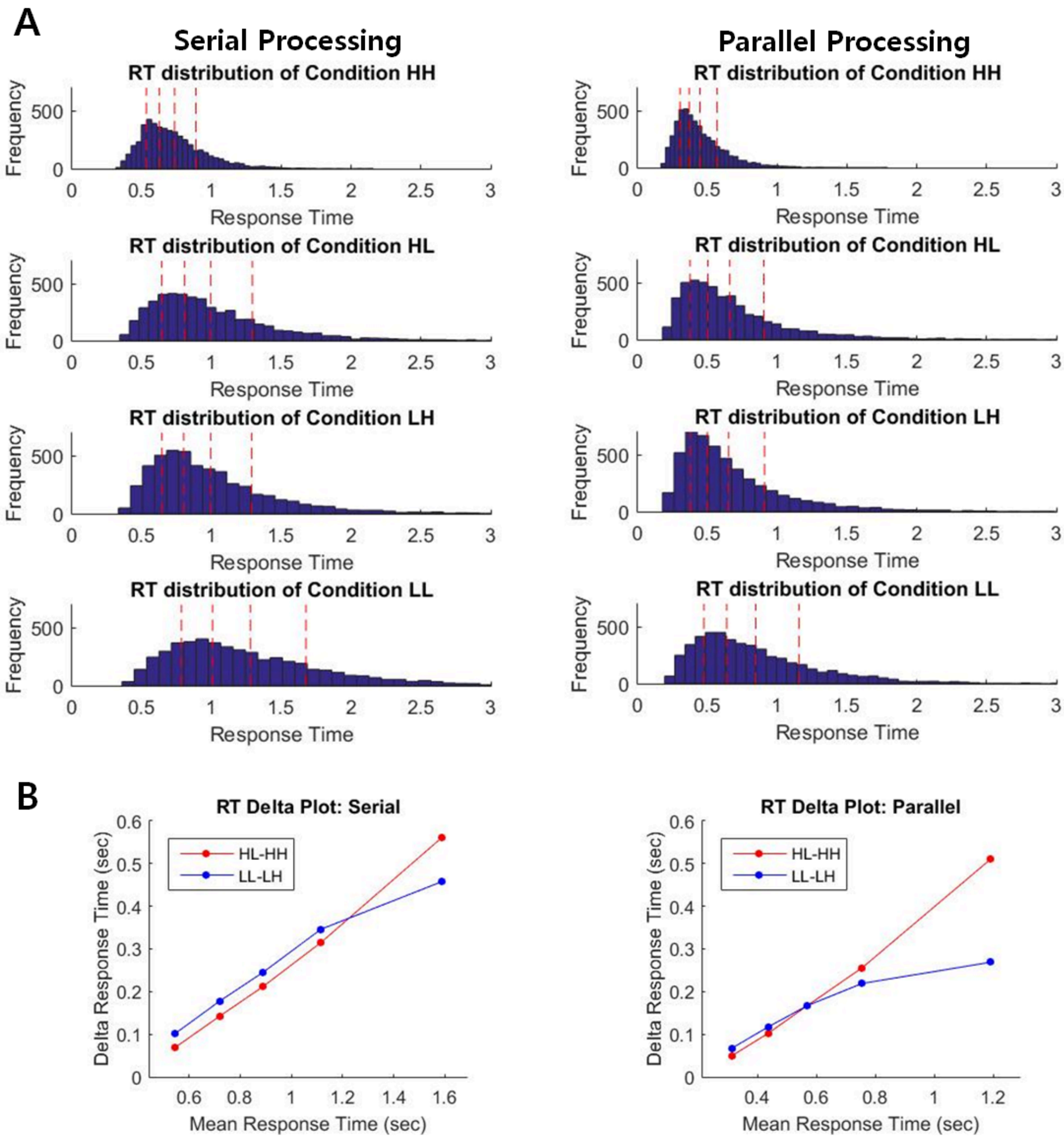

Figure 4. Ratcliff Diffusion Model Simulation
(A) RT distribution and its quintiles are shown for each coherence level combination.
(B) Medians were averaged between condition HL and LH for each quintile. The difference between the averaged medians and the medians of condition HH is shown with red line. The difference between the averaged medians and the medians of condition LL is shown with a blue line. True to our intuition, the difference is greater for parallel processing than for serial processing. The difference seems to exist mostly in the slower trials.

5.2 Experimental Performance

Similar analysis was conducted on the experimental subject data. As expected, accuracy was higher and response time was slower overall for subject set 2 than for subject set 1. For subject set 1, a total of 9122 trials were taken into account after excluding unreasonably quick (less than 0.2 sec) responses, of which 6469 were correctly responded. For subject set 2, a total of 9209 trials were taken into account after filtering quick responses, of which 7599 were correctly responded.

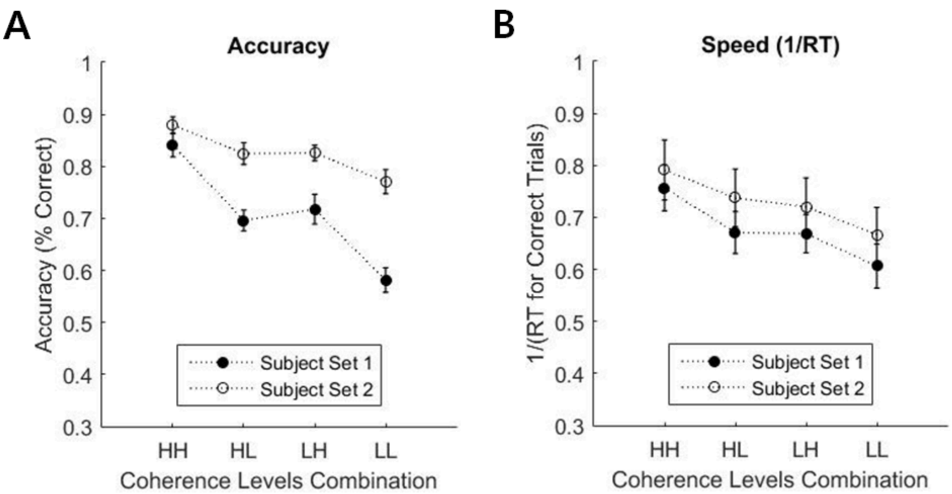

Figure 5. Subject Performance by Coherence Combination Conditions
(A) Accuracy
(B) Speed (1/RT)

Performing linear regression on each subject set revealed no significant interaction effect. Thus, evidence for parallel processing was not observed.

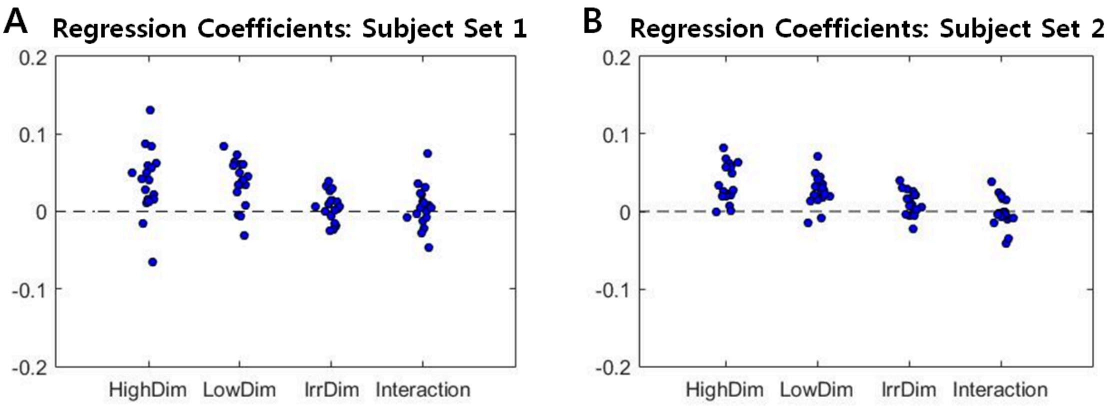

Figure 6. Regression Coefficients
Average speed coefficient was much higher in general, and was excluded from the above plots for easier comparison among the rest of the predictor variables.
(A) The coefficients for the irrelevant dimension's coherence level and interaction are distributed across the zero line, in congruence with our t-test results.
(B) The interaction variable's coefficients are distributed across the zero line, but the coefficients for the irrelevant dimension coherence level are in general scattered above the zero line.

For subject set 1, there was a significant effect of the average speed ($t(17) = 17.00$, $p<0.001$). There were also significant effects for the higher dimension feature's coherence level ($t(17)=3.78$, $p=0.002$) and the lower dimension feature's coherence level($t(17)=5.23$, $p<0.001$), with high coherence leading to higher speed for both predictor variables. As expected, the irrelevant dimension feature's coherence level did not exhibit a significant effect on performance speed ($t(17)=1.30$, $p=0.210$). Significant effect was also not observed for the interaction between the coherence levels of the two relevant dimensions ($t(17) = 0.90$, $p=0.383$).

For subject set 2, there was a significant effect of average speed ($t(17) = 12.98$, $p<0.001$). There were also significant effects for the higher dimension feature's coherence level ($t(17)=6.06$, $p<0.001$) and the lower dimension feature's coherence level($t(17)=5.69$, $p<0.001$), with high coherence leading to higher speed for both predictor variables. Unlike subject set 1, the irrelevant dimension feature's coherence level did exhibit significant effect on performance speed ($t(17)=2.99$, $p=0.008$). There was no effect of the interaction between the coherence levels of the two relevant dimensions ($t(17) = 0.01$, $p=0.99$). To examine whether the observed effect of the irrelevant dimension feature implies some parallel processing, we divided the trials depending on the coherence level of the higher dimension feature. We hypothesized that a subject may be more distractible when the decision-making for the higher dimension is more difficult. Regressions on the trials separated by the higher dimension feature's coherence level both revealed significant effect (High coherence: $t(17)=2.41$, $p=0.028$; Low coherence: $t(17)=2.34$, $p=0.032$), however, and no striking difference was observed.

We also looked for the interaction effect in each quintile of the RT distribution. In accordance to the regression result, no striking difference between the distances from conditions HH and LL to the averaged condition of HL and LH was observed for either subject set. Some interaction effect seemed to exist in the slowest quintile for both subject sets (See Figure 7B). The error bars are overlapped, however, and the within-subject comparison revealed no obvious trend for the interaction effect (Figure 7C).

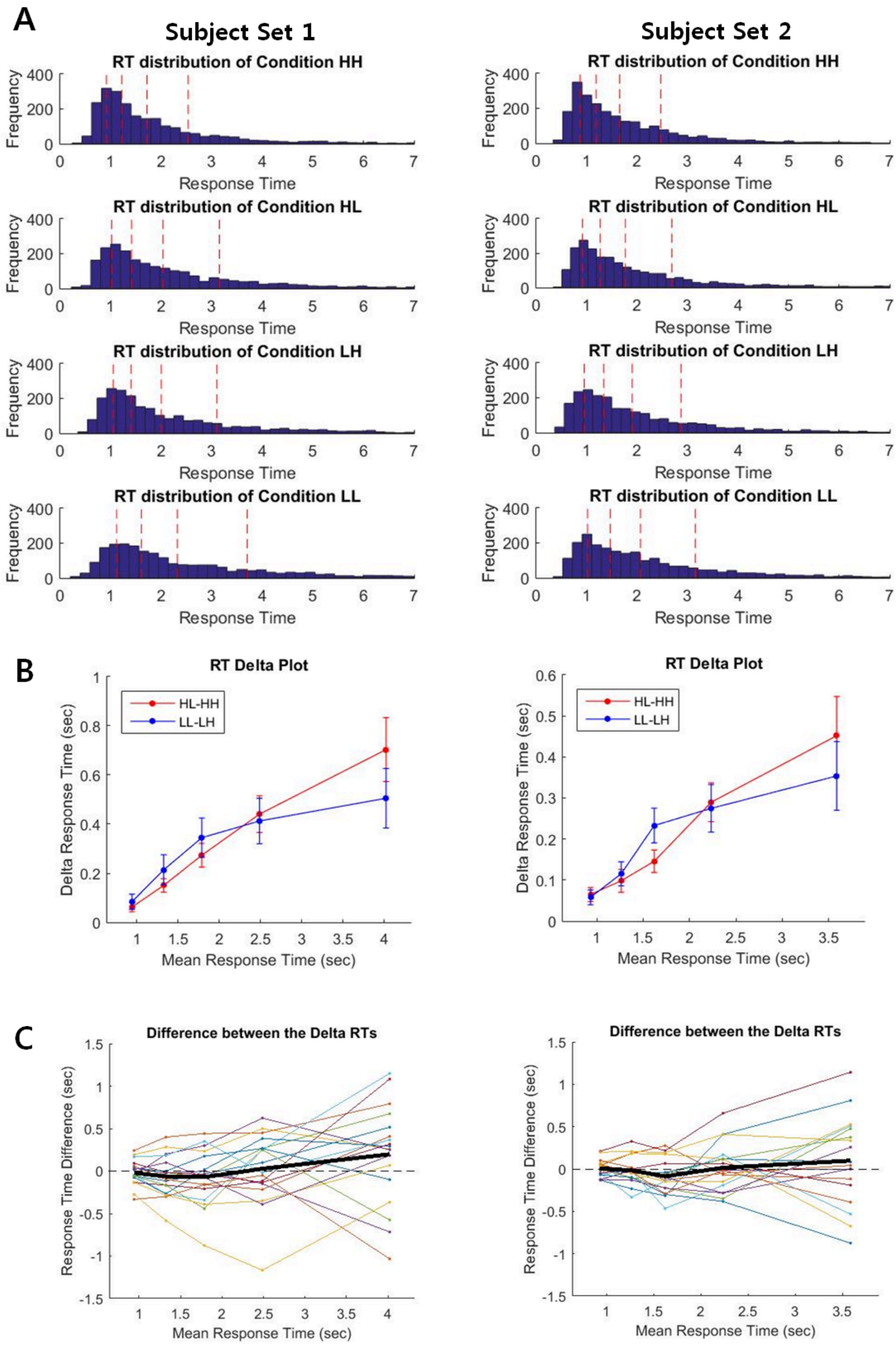
A
Subject Set 1
Subject Set 2
RT distribution of Condition HH
RT distribution of Condition HL
RT distribution of Condition LH
RT distribution of Condition LL
Frequency
Response Time
B
RT Delta Plot
HL-HH
LL-LH
Delta Response Time (sec)
Mean Response Time (sec)
C
Difference between the Delta RTs
Response Time Difference (sec)
Mean Response Time (sec)

Figure 7. Response Time by Quintile
(A) RT distribution and its quintiles are shown for each coherence level combination.
(B) Medians were averaged between condition HL and LH for each quintile. The difference between the averaged medians and the medians of condition HH and the medians of condition LL were computed separately for each subject. The mean and standard error of the computed differences are shown in the graph.
(C) The difference between the averaged medians and the medians of condition HH and the medians of condition LL were plotted for each subject. The thick black line describes the mean of the differences in each quintile.

5.3 Group Level Learning / Shift in Strategy

We examined the experimental data over time by applying the linear regression model to each of the four task blocks. We hypothesized that subjects may shift their strategy for processing information over time or with training. Second level regression analysis was performed with a linearly increasing predictor variable (-1.5:1:1.5). No significant linear relationship was observed for accuracy over blocks, while speed increased linearly over blocks with significance ($t(17)=4.89$, $p<0.0001$). No significant linear relationship was observed for the interaction coefficient from the first level regression analysis. Subject performance improved over time or with more practice in terms of response time, but subjects did not necessarily get any more accurate. Change in strategy from serial to parallel processing could be a plausible explanation, but such effect was not observed through the interaction term for either subject set.

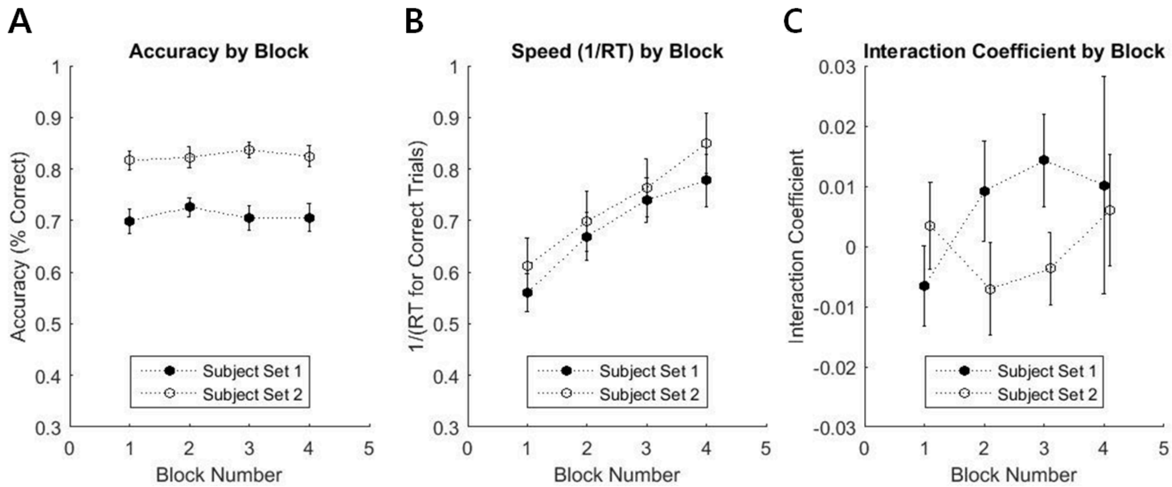

Figure 8. Analysis by Block
(A) Accuracy
(B) Speed
(C) Interaction Coefficients

5.4 Individual Differences in Strategy/ Learning of Strategy

Individual differences were also considered in interpreting the interaction effect and response time. For subject set 1, fitting a least-square regression line on a plot of average speed coefficient against the interaction term coefficient revealed a positive slope (Spearman's rank correlation: $r=0.3540$, $p=0.1499$) (Figure 9A). Subjects with a greater interaction effect tended to have greater speed. The analysis was repeated selectively for the trials in the slowest RT quartile, which strengthened the relationship (Spearman's rank correlation: $r=0.6574$, $p=0.0038$) (Figure 9B). A plot of the change in speed coefficient against the change in interaction coefficient also revealed a weak positive relationship (Spearman's rank correlation: $r=0.2301$, $p=0.3567$) (Figure 9C). Subjects with increasing interaction coefficient may be considered to be moving towards the parallel processing strategy.

For subject set 2, a plot of average speed coefficient against the interaction term coefficient did not exhibit a significant relationship (Spearman's rank correlation: $r = -0.0774$, $p=0.7607$) (Figure 9A). A plot of the change in speed coefficient against the change in interaction coefficient also did not reveal a significant relationship (Spearman's rank correlation: $r=0.1517$, $p=0.5467$) (Figure 9C).

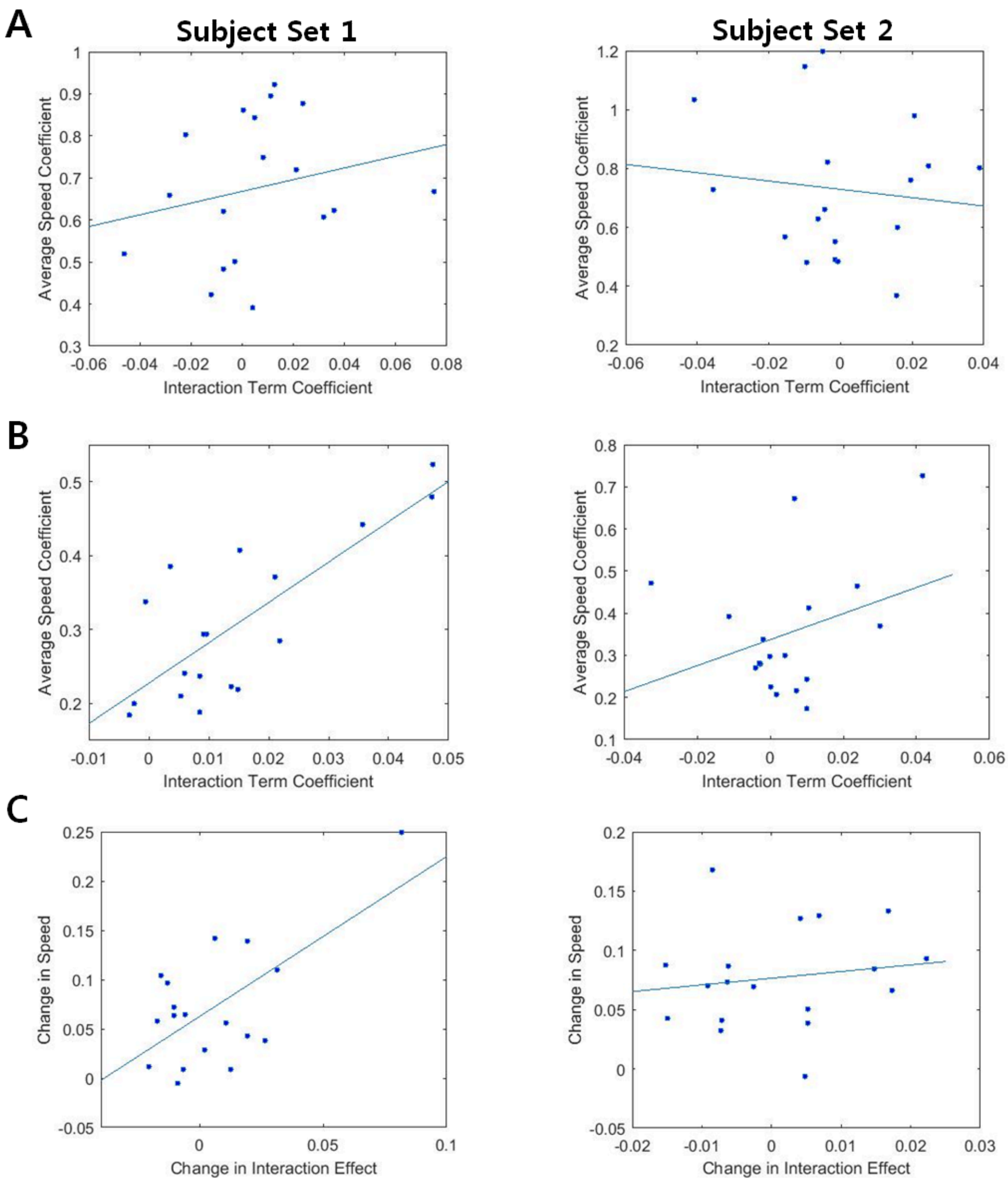

Figure 9. Individual Differences
(A) Interaction Term Coefficient vs. Average Speed Coefficient for all correct trials
(B) Interaction Term Coefficient vs. Average Speed Coefficient for correct trials in the slowest RT quartile. Subject set 1 showed strong positive correlation (Spearman's rank correlation: r=0.6574, p=0.0038), while subject set 2 showed weak positive correlation (Spearman's rank correlation: r=0.1269, p=0.6149)
(C) Change in Interaction effect vs. change in speed

6. Discussion

The present study aimed to develop a paradigm with which we could dissociate serial from parallel processing in a hierarchically-organized task. Our statistical approach focused on the “interaction effect” between the coherence levels of the high level dimension feature and the relevant lower level dimension feature as a key for distinguishing the two processing methods. Statistical analysis on simulated data did reveal a greater interaction effect for the parallel processing approach, thus serving as a basis for parameterizing the extent to which an individual’s strategy changes between serial and parallel processing methods. However, our analysis on human behavioral data revealed that although subject set 1 provided a trending evidence of possible change of strategy over time from serial to parallel processing, this observation was not consistent for subject set 2. We also did not observe the interaction effect for subjects in general, indicating that a parallel processing method may not be implemented for our task.

There are several potential reasons why our task failed to elicit the interaction effect from participants. Firstly, our task instructions were given to emphasize the hierarchical structure of our design, perhaps pushing the participants towards implementing the serial method. The instructions were delivered in such a manner to facilitate memorization of the task rules, which could be quite complicated if the hierarchical structure was unknown to begin with. The hierarchical design itself also presented a slight advantage for the serial method in terms of efficiency, as the parallel method would involve processing extra information on an unnecessary feature.

It also came to our attention that the deadline or dynamic urgency signal also may significantly affect the decision making process (Cisek et al. 2009, Hawkins et al. 2015) and thus could result in the similar effect as our hypothesized parallel processing model. The dynamic urgency signal is triggered when participants sacrifice accuracy for speed selectively in hard trials. In such cases, instead of waiting for the accumulated evidence to reach a fixed threshold for a hypothesis, the separation boundaries of DDM would collapse, terminating the decision-making process prematurely. Since this happens selectively for the harder (and slower) trials, the response time of the trials in condition LL would decrease, reducing the difference of response time with conditions LH and HL. So the dynamic urgency signal could be an alternative explanation for the "interaction effect" if it were to be observed in our experimental data. An urgency signal would, however, also predict greater error rate for the slower trials, as the decision-making process is terminated before collecting enough evidence to reach the normal decision threshold. Thus, comparing error rates between the faster trials and the slower trials would allow us to dissociate the urgency signal from the parallel processing model. For our experimental data, accuracy did turn out to be worst for the slowest RT quartile for both subject sets (Subject set 1: 0.68, 0.78, 0.76, 0.66; Subject set 2: 0.82, 0.85, 0.85, 0.80; in the order of the fastest to the slowest quartile). However, accuracy for the fastest RT quartile was also observed to be only slightly better, and further analysis would be needed to determine what caused such trend in accuracy. For future studies, we could simulate data with dynamic urgency signal and see whether it indeed loads on the interaction term the way we expect. Then we could individually fit the two models—urgency signal and parallel—to the experimental data and observe the better fit. Also, as no explicit evidence was found for parallel processing in this study, we could manipulate the task design to check for the evidence of serial processing instead.

7 References

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8 Appendix: Mathematical Proof of Intuition

Let X be a random variable whose associated distribution describes the reaction time of a binary decision with a high coherence stimulus. Let Y be another random variable such that YX is a random variable whose associated distribution describes the reaction time of a binary decision with a low coherence stimulus. Because reaction time is always positive and a lower coherence level increases the difficulty of making decision, we assume $X > 0$ and $Y > 1$.

There are two different coherence levels for each of the two dimension features relevant to our decision making, and thus there exist four possible combinations of coherence levels in total. For each combination, the reaction time according to the serial or parallel processing model can be expressed as a function of X and Y as follows:

	High Dim Lev & Low Dim Lev	Serial RT	Parallel RT
(A)	High coh (High coh (	→	
(B)	High coh (Low coh (	→	
(C)	Low coh (High coh (	→	
(D)	Low coh (Low coh (	→	

We'd like to show that $(B) - (A) = (D) - (C)$ is true in average for the reaction time according to the serial processing model and that $(B) - (A) > (D) - (C)$ is true in average for the reaction time according to the parallel processing model.

Let us first look at the serial processing model. We want to prove:

$$E\left[X_3 + Y_4X_4\right] - E\left[X_1 + X_2\right] = E\left[Y_7X_7 + Y_8X_8\right] - E\left[Y_5X_5 + X_6\right]$$

where each X_i is a random variable with identical distribution as X, and each X_iY_i is a random variable with identical distribution as XY. Then, using the basic property of expected value, we can reduce down the above equation as follows:

$$E[X_3] + E[Y_4X_4] - (E[X_1] + E[X_2]) = E[Y_7X_7] + E[Y_8X_8] - (E[Y_5X_5] + E[X_6])$$
$$E[X] + E[YX] - E[X] - E[X] = E[YX] + E[YX] - E[YX] - E[X]$$
$$E[YX] - E[X] = E[YX] - E[X]$$

The equality holds and thus we have demonstrated that there is no "interaction effect" for the serial processing model.

For the parallel processing model, we want to prove:

$$E[(X_3, Y_4X_4)] - E[(X_1, X_2)] > E[(Y_7X_7, Y_8X_8)] - E[(Y_5X_5, X_6)]$$

where each X_i is a random variable with identical distribution as X, and each X_iY_i is a random variable with identical distribution as XY. Using the basic property of expected value, this can also be rewritten as:

$$E[(X_1, Y_2X_2) - (X_1, X_2) - (Y_1X_1, Y_2X_2) + (Y_1X_1, X_2)] > 0$$

Let us first consider all the possible orderings of the four variables involved: X_1, X_2, Y_1X_1, and Y_2X_2. Because we have $Y_1,\ Y_2 > 1$, we can rule out all the orderings that contradict $X_1 < Y_1X_1$ or $X_2 < Y_2X_2$. We end up with the following 6 orderings:

(a) (d)
(b) (e)
(c) (f)

We can split up each ordering into more specific cases and compute the left hand side of the previous inequality under each condition:

Ordering	Specific Cases	The Inequality Reduces to...
(a)	$X_1 < X_2 < Y_1X_1 < Y_2X_2$ $X_1 < X_2 < Y_1X_1 = Y_2X_2$ $X_1 < X_2 = Y_1X_1 < Y_2X_2$ $X_1 = X_2 < Y_1X_1 < Y_2X_2$ $X_1 = X_2 < Y_1X_1 = Y_2X_2$	$Y_1X_1 - X_2 > 0$ $Y_1X_1 - X_2 > 0$ $Y_1X_1 - X_2 = 0$ $Y_1X_1 - X_2 > 0$ $Y_1X_1 - X_2 > 0$
(b)	$X_1 < X_2 < Y_2X_2 < Y_1X_1$ $X_1 < X_2 < Y_2X_2 = Y_1X_1$ $X_1 = X_2 < Y_2X_2 < Y_1X_1$ $X_1 = X_2 < Y_2X_2 = Y_1X_1$	$Y_2X_2 - X_2 > 0$ $Y_2X_2 - X_2 > 0$ $Y_2X_2 - X_2 > 0$ $Y_2X_2 - X_2 > 0$
(c)	$X_1 < Y_1X_1 < X_2 < Y_2X_2$ $X_1 < Y_1X_1 = X_2 < Y_2X_2$	$Y_2X_2 - X_2 - \left(Y_2X_2 - X_2\right) = 0$ $Y_2X_2 - X_2 - \left(Y_2X_2 - X_2\right) = 0$
(d)	$X_2 < X_1 < Y_1X_1 < Y_2X_2$ $X_2 < X_1 < Y_1X_1 = Y_2X_2$	$Y_1X_1 - X_1 > 0$ $Y_1X_1 - X_1 > 0$
(e)	$X_2 < X_1 < Y_2X_2 < Y_1X_1$ $X_2 < X_1 < Y_2X_2 = Y_1X_1$ $X_2 < X_1 = Y_2X_2 < Y_1X_1$	$Y_2X_2 - X_1 > 0$ $Y_2X_2 - X_1 > 0$ $Y_2X_2 - X_1 = 0$
(f)	$X_2 < Y_2X_2 < X_1 < Y_1X_1$ $X_2 < Y_2X_2 = X_1 < Y_1X_1$	$X_1 - X_1 - \left(Y_1X_1 - Y_1X_1\right) = 0$ $X_1 - X_1 - \left(Y_1X_1 - Y_1X_1\right) = 0$

Since $\left(X_1, Y_2X_2\right) - \left(X_1, X_2\right) - \left(Y_1X_1, Y_2X_2\right) + \left(Y_1X_1, X_2\right)$ is greater than or equal to 0 for each of the specific cases, its expected value overall must be greater than zero. Thus, the original inequality holds, and we have demonstrated that there exists "interaction effect" for the parallel processing mode.